

could not have been guessed from its worst predecessor, a one-day loss of around 10 percent—and this applies to almost all such events, of course. My results were that regular events can predict regular events, but that extreme events, perhaps because they are more acute when people are unprepared, are almost never predicted from narrow reliance on the past.

The fact that this notion is not obvious to people is shocking to me. It is particularly shocking that people do what are called “stress tests” by taking the worst possible *past* deviation as an anchor event to project the worst possible future deviation, not thinking that they would have failed to account for that past deviation had they used the same method on the day before the occurrence of that past anchor event.*

These people have PhDs in economics; some are professors—one of them is the chairman of the Federal Reserve (at the time of writing). Do advanced degrees make people blind to these elementary notions?

Indeed, the Latin poet Lucretius, who did not attend business school, wrote that we consider the biggest object of any kind that we have seen in our lives as the largest possible item: *et omnia de genere omni / Maxima quae vivit quisque, haec ingentia fingit.*

PROBABILITY HAS TO BE SUBJECTIVE†

This raises a problem that is worth probing in some depth. The fact that many researchers do not realize immediately that the Black Swan corresponds mainly to an incomplete map of the world, or that some researchers have to stress this subjective quality (Jochen Runde, for instance, wrote an insightful essay on the Black Swan idea, but one in which he felt he needed to go out of his way to stress its subjective aspect), takes us to the historical problem in the very definition of probability. Historically, there have been many approaches to the philosophy of probability. The notion that two people can have two different views of the world, then express them as different probabilities remained foreign to the research. So it took a while for scientific researchers to accept the non-Asperger notion that different people can, while being rational, assign different probabilities to different future states of the world. This is called “subjective probability.”

Subjective probability was formulated by Frank Plumpton Ramsey in 1925 and Bruno de Finetti in 1937. The take on probability by these two intellectual giants is that it can be represented as a quantification of the degree of belief (you set a number between 0 and 1 that corresponds to